

Project Testing Phase – EduGenie

1. Introduction

The Project Testing Phase is the stage where the EduGenie application is checked to identify errors, bugs, and performance issues. Testing ensures that all features work correctly and meet the project requirements.

2. Objectives of Testing

To identify and fix errors in the application.

To verify that all modules work correctly.

To check the accuracy of AI-based features.

To ensure data is stored and retrieved properly.

To provide a secure and user-friendly application.

3. Types of Testing

1. Unit Testing

Each individual module is tested separately, such as login, quiz, AI assistant, and progress tracking.

2. Integration Testing

Different modules are tested together to ensure they communicate and work properly.

3. Functional Testing

The functions of EduGenie are tested according to the requirements.

4. UI Testing

The user interface is checked for proper layout, buttons, navigation, and readability.

5. Database Testing

Student details, quiz results, study materials, and progress data are checked for correct storage and retrieval.

6. AI Testing

The AI assistant is tested to check whether it provides relevant and appropriate responses to student queries.

7. Performance Testing

The application is tested for speed, response time, and stability when multiple users access it.

8. Security Testing

Login, user data, and database access are tested to ensure that information is protected.

4. Testing Process

1. Preparing test cases.
2. Testing each module.
3. Identifying errors and bugs.
4. Fixing the identified issues.
5. Re-testing the corrected modules.

6. Performing final system testing.
7. Confirming that the application is ready for deployment.

5. Test Cases

Test Case	Expected Result
User Login	User should successfully log in with valid details
Registration	New user should be registered successfully
AI Assistant	AI should provide relevant responses
Quiz	Questions and answers should work correctly
Score Calculation	Score should be calculated accurately
Progress Tracking	Student progress should be updated correctly
Database	Data should be stored and retrieved correctly
Navigation	All pages should open correctly

6. Bug Fixing

When an error is found, it is recorded and analyzed. Developers correct the problem and perform testing again to ensure that the issue has been resolved.

7. Expected Outcome

After testing, EduGenie should be:

Error-free as far as practical.

Reliable and stable.

Easy to use.

Secure.

Accurate in its major functions.

Ready for deployment.

8. Conclusion

The Project Testing Phase ensures that EduGenie works according to its requirements. Proper testing and bug fixing improve the quality, reliability, security, and usability of the educational platform.